

Matrices, Vectors & Complex Numbers

ANSWER KEY



Matrix arithmetic

- 1 Matrix A has rows $(2, 3)$ and $(-1, 4)$; matrix B has rows $(5, -2)$ and $(1, 3)$. Find the rows of $A + B$.

Adding corresponding entries: first row $(2 + 5, 3 + (-2)) = (7, 1)$; second row $(-1 + 1, 4 + 3) = (0, 7)$.

- 2 Find $3A$ for matrix A with rows $(1, -2)$ and $(4, 0)$.

Multiply every entry by 3: rows $(3, -6)$ and $(12, 0)$.

Matrix multiplication

- 3 Matrix A has rows $(1, 2)$ and $(3, 4)$; matrix B has rows $(5, 6)$ and $(7, 8)$. Find AB .

Row 1 of AB : $(1(5) + 2(7), 1(6) + 2(8)) = (19, 22)$. Row 2 of AB :
 $(3(5) + 4(7), 3(6) + 4(8)) = (43, 50)$.

Determinants and inverses of 2×2 matrices

- 4 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 5)$.

$$\det = 4(5) - 2(3) = 20 - 6 = 14.$$

- 5 Find the inverse of the matrix M with rows $(2, 1)$ and $(5, 3)$.

$\det(M) = 2(3) - 1(5) = 1$. $M^{-1} = \frac{1}{1} \times$ the matrix with rows $(3, -1)$ and $(-5, 2)$, giving M^{-1} with rows $(3, -1)$ and $(-5, 2)$.

Vectors: magnitude and direction

- 6 Find the magnitude of the vector $\mathbf{v} = (3, 4)$.

$$|\mathbf{v}| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

- 7 Given $\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-3, 7)$, find $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{b}$.

$$\mathbf{a} + \mathbf{b} = (2 - 3, -5 + 7) = (-1, 2). \quad 2\mathbf{a} - \mathbf{b} = (4 - (-3), -10 - 7) = (7, -17).$$

The dot product

- 8 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (-1, 4)$, and determine whether the vectors are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 2(4) = -3 + 8 = 5$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, the vectors are not perpendicular.

Complex number arithmetic

- 9 Simplify $(4 + 3i) + (2 - 5i)$.

$$(4 + 2) + (3 - 5)i = 6 - 2i.$$

- 10 Simplify $(3 + 2i)(1 - 4i)$.

$$(3 + 2i)(1 - 4i) = 3 - 12i + 2i - 8i^2 = 3 - 10i + 8 = 11 - 10i \text{ (using } i^2 = -1\text{)}.$$

Complex conjugates and division

- 11 Find the complex conjugate of $z = 6 - 5i$, and compute $z\bar{z}$.

$$\bar{z} = 6 + 5i. \quad z\bar{z} = (6 - 5i)(6 + 5i) = 36 - 25i^2 = 36 + 25 = 61.$$

- 12 Simplify $\frac{5+i}{2-i}$ by multiplying by the conjugate of the denominator.

$$\frac{5+i}{2-i} \cdot \frac{2+i}{2+i} = \frac{(5+i)(2+i)}{4+1} = \frac{10+5i+2i+i^2}{5} = \frac{10+7i-1}{5} = \frac{9+7i}{5} = \frac{9}{5} + \frac{7}{5}i.$$

Solving quadratics with complex solutions

- 13 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

$$\text{Using the quadratic formula: } x = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i.$$

Combining matrices and systems of equations

- 14 This question has two parts. The system $3x + 2y = 16$ and $x + 4y = 18$ can be written as a matrix equation $M\mathbf{v} = \mathbf{c}$, where M has rows $(3, 2)$ and $(1, 4)$. (a) Find $\det(M)$ and M^{-1} . (b) Use M^{-1} to solve the system for x and y .

(a) $\det(M) = 3(4) - 2(1) = 10$. $M^{-1} = \frac{1}{10} \times$ the matrix with rows $(4, -2)$ and $(-1, 3)$, giving rows $(0.4, -0.2)$ and $(-0.1, 0.3)$. (b) Applying M^{-1} to $(16, 18)^T$: $x = 0.4(16) - 0.2(18) = 6.4 - 3.6 = 2.8$.
 $y = -0.1(16) + 0.3(18) = -1.6 + 5.4 = 3.8$.

Powers of i and simplifying complex expressions

- 15 Simplify i^{17} , using the fact that powers of i cycle with period 4.

$$17 = 4(4) + 1, \text{ so } i^{17} = (i^4)^4 \cdot i^1 = 1^4 \cdot i = i.$$

- 16 Simplify $(2i)^3$.

$$(2i)^3 = 8i^3 = 8(i^2 \cdot i) = 8(-1 \cdot i) = -8i.$$

Vector direction and unit vectors

- 17 Find a unit vector in the same direction as $\mathbf{v} = (6, -8)$.

$$|\mathbf{v}| = \sqrt{36 + 64} = \sqrt{100} = 10. \text{ Unit vector: } \frac{1}{10}(6, -8) = (0.6, -0.8).$$

- 18 Find the angle a vector $\mathbf{v} = (3, 3)$ makes with the positive x-axis.

$$\theta = \arctan\left(\frac{3}{3}\right) = \arctan(1) = 45^\circ.$$

Identity matrix and matrix equations

- 19 Explain why $MI = M$ for any matrix M and identity matrix I of matching size, and verify this for M with rows $(2, 3)$ and $(1, 4)$ multiplied by the 2×2 identity matrix (rows $(1, 0)$ and $(0, 1)$).

The identity matrix leaves any matrix unchanged under multiplication, just as the number 1 does in ordinary multiplication. Verifying: row 1 of MI : $(2(1) + 3(0), 2(0) + 3(1)) = (2, 3)$; row 2: $(1(1) + 4(0), 1(0) + 4(1)) = (1, 4)$ — identical to M 's original rows, confirming $MI = M$.

A multi-part complex number problem

- 20 This question has two parts. Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. (a) Find $|z_1|$ (the modulus of z_1). (b) Find $z_1 \cdot z_2$ and its modulus, then verify that $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$.

$$\begin{aligned} \text{(a) } |z_1| &= \sqrt{3^2 + 4^2} = \sqrt{25} = 5. \text{ (b) } z_1 z_2 = (3 + 4i)(1 - 2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i. \\ |z_1 z_2| &= \sqrt{11^2 + 2^2} = \sqrt{121 + 4} = \sqrt{125} = 5\sqrt{5}. |z_2| = \sqrt{1^2 + (-2)^2} = \sqrt{5}. \text{ Check: } |z_1| \cdot |z_2| = 5\sqrt{5}, \\ &\text{matching } |z_1 z_2| = 5\sqrt{5} \checkmark. \end{aligned}$$

Scalar multiplication of matrices and vectors combined

- 21 Given vectors $\mathbf{u} = (4, -3)$ and $\mathbf{v} = (-2, 5)$, find $3\mathbf{u} - 2\mathbf{v}$.

$$3\mathbf{u} = (12, -9). 2\mathbf{v} = (-4, 10). 3\mathbf{u} - 2\mathbf{v} = (12 - (-4), -9 - 10) = (16, -19).$$

Multiple choice — easy

22 Simplify $(10) + (4 - 2i)$.

- a. 14
- b. $-14 - 2i$
- c. $14 - 2i$ ✓
- d. $16 - 2i$

$$(10 + 4) + (0 - 2)i = 14 - 2i.$$

23 Simplify $(-10 + 6i) - (7 + 11i)$.

- a. $-17 - 5i$ ✓
- b. $-17 - 3i$
- c. $-15 - 5i$
- d. $17 - 5i$

$$(-10 - 7) + (6 - 11)i = -17 - 5i.$$

24 Simplify $(3 + 9i) + (11 + 9i)$.

- a. $-14 + 18i$
- b. $14 + 20i$
- c. $16 + 18i$
- d. $14 + 18i$ ✓

$$(3 + 11) + (9 + 9)i = 14 + 18i.$$

25 Simplify $(2 - 9i) - (1 + 7i)$.

a. $3 - 16i$

b. $1 - 16i$ ✓

c. $-1 - 16i$

d. $1 - 14i$

$$(2 - 1) + (-9 - 7)i = 1 - 16i.$$

26 Simplify $(6 - 10i) + (-9 + 8i)$.

a. $-1 - 2i$

b. $3 - 2i$

c. $-3 - 2i$ ✓

d. -3

$$(6 - 9) + (-10 + 8)i = -3 - 2i.$$

27 Simplify $(-4 + 8i) - (-10 - 3i)$.

a. $8 + 11i$

b. $6 + 11i$ ✓

c. $-6 + 11i$

d. $6 + 13i$

$$(-4 + 10) + (8 + 3)i = 6 + 11i.$$

28 Simplify $(-5 + 5i) - (9 - 10i)$.

a. $-14 + 17i$

b. $-14 + 15i$ ✓

c. $-12 + 15i$

d. $14 + 15i$

$$(-5 - 9) + (5 + 10)i = -14 + 15i.$$

29 Simplify $(1 - 12i) + (8 + 5i)$.

a. $11 - 7i$

b. $-9 - 7i$

c. $9 - 7i$ ✓

d. $9 - 5i$

$$(1 + 8) + (-12 + 5)i = 9 - 7i.$$

30 Simplify $(-9 - 12i) - (-12i)$.

a. -7

b. $-9 + 2i$

c. -9 ✓

d. 9

$$(-9) + (-12 + 12)i = -9.$$

31 Simplify $(-1 + 3i) - (11 + 5i)$.

a. $-10 - 2i$

b. -12

c. $12 - 2i$

d. $-12 - 2i$ ✓

$$(-1 - 11) + (3 - 5)i = -12 - 2i.$$

32 Simplify $(12 - 6i) + (-3 + 5i)$.

a. $9 - i$ ✓

b. $9 + i$

c. $11 - i$

d. $-9 - i$

$$(12 - 3) + (-6 + 5)i = 9 - i.$$

33 Simplify $(-11 + 12i) + (8 + 11i)$.

a. $3 + 23i$

b. $-3 + 25i$

c. $-1 + 23i$

d. $-3 + 23i$ ✓

$$(-11 + 8) + (12 + 11)i = -3 + 23i.$$

34 Simplify $(-8 + 10i) + (-9i)$.

a. $-8 + i$ ✓

b. $-8 + 3i$

c. $-6 + i$

d. $8 + i$

$$(-8) + (10 - 9)i = -8 + i.$$

35 Simplify $(2 - i) + (-7 + 4i)$.

a. $5 + 3i$

b. $-5 + 5i$

c. $-3 + 3i$

d. $-5 + 3i$ ✓

$$(2 - 7) + (-1 + 4)i = -5 + 3i.$$

36 Simplify $(-12i) + (3 - 6i)$.

a. $-3 - 18i$

b. $5 - 18i$

c. $3 - 16i$

d. $3 - 18i$ ✓

$$(0 + 3) + (-12 - 6)i = 3 - 18i.$$

Multiple choice — medium

37 Simplify $(2 - 7i)(3 - 9i)$.

a. $-57 - 39i$ ✓

b. $-55 - 39i$

c. $69 - 39i$

d. $-57 + 3i$

$$(2 - 7i)(3 - 9i) = 6 - 21i - 18i + 63i^2 = 6 - 39i - 63 = -57 - 39i.$$

38 Simplify $(-4 - 4i)(8 - 9i)$.

a. $-66 + 4i$

b. $4 + 4i$

c. $-68 + 4i$ ✓

d. $-68 + 68i$

$$(-4 - 4i)(8 - 9i) = -32 - 32i + 36i + 36i^2 = -32 + 4i - 36 = -68 + 4i.$$

39 Simplify $(-4 + 2i)(7 - 9i)$.

- a. $-46 + 50i$
- b. $-10 + 22i$
- c. $-10 + 50i$ ✓
- d. $-8 + 50i$

$$(-4 + 2i)(7 - 9i) = -28 + 14i + 36i - 18i^2 = -28 + 50i + 18 = -10 + 50i.$$

40 Simplify $(7 + i)(7 + 9i)$.

- a. $42 + 70i$
- b. $40 + 70i$ ✓
- c. $40 + 56i$
- d. $58 + 70i$

$$(7 + i)(7 + 9i) = 49 + 7i + 63i + 9i^2 = 49 + 70i - 9 = 40 + 70i.$$

41 Simplify $(-3 + i)(-2 - 3i)$.

- a. $11 + 7i$
- b. $9 + 11i$
- c. $9 + 7i$ ✓
- d. $3 + 7i$

$$(-3 + i)(-2 - 3i) = 6 - 2i + 9i - 3i^2 = 6 + 7i + 3 = 9 + 7i.$$

42 Simplify $(3 + 7i)(-5 - 6i)$.

- a. $27 + 17i$
- b. $27 - 53i$ ✓
- c. $-57 - 53i$
- d. $29 - 53i$

$$(3 + 7i)(-5 - 6i) = -15 - 35i - 18i - 42i^2 = -15 - 53i + 42 = 27 - 53i.$$

43 Simplify $(5i)(2 + 2i)$.

- a. $10 + 10i$
- b. $-10 - 10i$
- c. $-10 + 10i$ ✓
- d. $-8 + 10i$

$$(5i)(2 + 2i) = 0 + 10i + 10i^2 = 0 + 10i - 10 = -10 + 10i.$$

44 Simplify $(-5 + 3i)(9 + 6i)$.

- a. $-27 - 3i$
- b. $-61 - 3i$
- c. $-63 - 57i$
- d. $-63 - 3i$ ✓

$$(-5 + 3i)(9 + 6i) = -45 + 27i - 30i + 18i^2 = -45 - 3i - 18 = -63 - 3i.$$

45 Simplify $(2 - 7i)(-7 - 5i)$.

- a. $-47 + 39i$
- b. $21 + 39i$
- c. $-49 + 39i$ ✓
- d. $-49 - 59i$

$$(2 - 7i)(-7 - 5i) = -14 + 49i - 10i + 35i^2 = -14 + 39i - 35 = -49 + 39i.$$

46 Simplify $(2 - 3i)(3 + 3i)$.

- a. $15 + 15i$
- b. $15 - 3i$ ✓
- c. $17 - 3i$
- d. $-3 - 3i$

$$(2 - 3i)(3 + 3i) = 6 - 9i + 6i - 9i^2 = 6 - 3i + 9 = 15 - 3i.$$

47 Simplify $(8 - 2i)(-6 - 4i)$.

- a. $-54 - 20i$
- b. $-56 - 20i$ ✓
- c. $-40 - 20i$
- d. $-56 - 44i$

$$(8 - 2i)(-6 - 4i) = -48 + 12i - 32i + 8i^2 = -48 - 20i - 8 = -56 - 20i.$$

48 Simplify $(6 + 3i)(8 - 6i)$.

- a. $66 - 60i$
- b. $66 - 12i$ ✓
- c. $30 - 12i$
- d. $68 - 12i$

$$(6 + 3i)(8 - 6i) = 48 + 24i - 36i - 18i^2 = 48 - 12i + 18 = 66 - 12i.$$

49 Simplify $(-3 + 6i)(4 - 2i)$.

- a. $30i$ ✓
- b. $-24 + 30i$
- c. $2 + 30i$
- d. $-18i$

$$(-3 + 6i)(4 - 2i) = -12 + 24i + 6i - 12i^2 = -12 + 30i + 12 = 30i.$$

50 Simplify $(-8 + 4i)(-2 - 9i)$.

a. $-20 + 64i$

b. $54 + 64i$

c. $52 + 64i$ ✓

d. $52 + 80i$

$$(-8 + 4i)(-2 - 9i) = 16 - 8i + 72i - 36i^2 = 16 + 64i + 36 = 52 + 64i.$$

51 Simplify $(-6 + 8i)(-7 + 9i)$.

a. $-28 - 110i$

b. $-30 + 2i$

c. $-30 - 110i$ ✓

d. $114 - 110i$

$$(-6 + 8i)(-7 + 9i) = 42 - 56i - 54i + 72i^2 = 42 - 110i - 72 = -30 - 110i.$$

Multiple choice — hard

52 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 1)$.

a. -4

b. 0

c. -2 ✓

d. 10

$$\det = (4)(1) - (2)(3) = 4 - 6 = -2.$$

53 Find the determinant of the matrix with rows $(-4, 5)$ and $(7, -5)$.

- a. -13
- b. -17
- c. -3
- d. -15 ✓

$$\det = (-4)(-5) - (5)(7) = 20 - 35 = -15.$$

54 Find the determinant of the matrix with rows $(-1, -7)$ and $(-2, 4)$.

- a. -20
- b. -18 ✓
- c. 30
- d. -16

$$\det = (-1)(4) - (-7)(-2) = -4 - 14 = -18.$$

55 Find the determinant of the matrix with rows $(-8, -9)$ and $(9, 8)$.

- a. 17 ✓
- b. 15
- c. 19
- d. 0

$$\det = (-8)(8) - (-9)(9) = -64 + 81 = 17.$$

56 Find the determinant of the matrix with rows $(-7, 6)$ and $(8, 6)$.

- a. -92
- b. -88
- c. -90 ✓
- d. -86

$$\det = (-7)(6) - (6)(8) = -42 - 48 = -90.$$

57 Find the magnitude of the vector $\mathbf{v} = (6, 8)$.

a. 10 ✓

b. 14

c. 9

d. 11

$$|\mathbf{v}| = \sqrt{(6)^2 + (8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

58 Find the magnitude of the vector $\mathbf{v} = (20, 21)$.

a. 29 ✓

b. 41

c. 30

d. 28

$$|\mathbf{v}| = \sqrt{(20)^2 + (21)^2} = \sqrt{400 + 441} = \sqrt{841} = 29.$$

59 Find the magnitude of the vector $\mathbf{v} = (-3, 4)$.

a. 4

b. 7

c. 5 ✓

d. 6

$$|\mathbf{v}| = \sqrt{(-3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

60 Find the magnitude of the vector $\mathbf{v} = (20, -21)$.

a. 28

b. 29 ✓

c. 41

d. 30

$$|\mathbf{v}| = \sqrt{(20)^2 + (-21)^2} = \sqrt{400 + 441} = \sqrt{841} = 29.$$

61 Find the magnitude of the vector $\mathbf{v} = (-7, -24)$.

a. 26

b. 24

c. 25 ✓

d. 31

$$|\mathbf{v}| = \sqrt{(-7)^2 + (-24)^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

62 Simplify i^{16} , using the fact that powers of i cycle with period 4.

a. i

b. $-i$

c. 1 ✓

d. -1

$$16 = 4(4), \text{ so } i^{16} = (i^4)^4 = 1^4 = 1.$$

63 Simplify i^{36} , using the fact that powers of i cycle with period 4.

a. 1 ✓

b. $-i$

c. i

d. -1

$$36 = 4(9), \text{ so } i^{36} = (i^4)^9 = 1^9 = 1.$$

64 Simplify i^5 , using the fact that powers of i cycle with period 4.

a. -1

b. 1

c. $-i$

d. i ✓

$$5 = 4(1) + 1, \text{ so } i^5 = (i^4)^1 \cdot i^1 = 1^1 \cdot i^1 = i.$$

65 Simplify i^{40} , using the fact that powers of i cycle with period 4.

a. -1

b. i

c. $-i$

d. 1 ✓

$$40 = 4(10), \text{ so } i^{40} = (i^4)^{10} = 1^{10} = 1.$$

66 Simplify i^{34} , using the fact that powers of i cycle with period 4.

a. 1

b. $-i$

c. i

d. -1 ✓

$$34 = 4(8) + 2, \text{ so } i^{34} = (i^4)^8 \cdot i^2 = 1^8 \cdot i^2 = -1.$$